

Predicting News Publishers From Headlines Using Deep Learning

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Introduction

News headlines are often shared independently of their original sources, especially on social media platforms. As a result, readers may consume information without knowing which publication produced it. Additionally, while many news outlets cover the same stories, each publication has different underlying stylistic differences in their coverage that readers may not pick up on. This project explores whether those differences are strong enough to be detected automatically using machine learning techniques.

The goal of this project is to predict the publishing company of a news article using only the headline text. By focusing solely on headlines, we remove contextual information from article bodies and examine whether short-form text contains enough stylistic or linguistic signals to distinguish publishers. This task has potential applications in media literacy, source attribution, and misinformation detection, where identifying the origin of a headline could provide important context to readers.

Problem Description

The central problem addressed in this project is multi-class text classification, where each headline must be assigned to one of ten possible news publishers. The challenge lies in the short length of headlines and the fact that many outlets cover the same events using similar wording.

Our research question was:

- Do deep learning models outperform simpler text-based models when classifying news publishers using only headline text?

Our original hypothesis was that more complex deep learning architectures would outperform simpler models. Specifically, we expected convolutional neural networks (CNNs) to perform best, followed by long short-term memory networks (LSTMs), with feed-forward networks performing the worst. However, as results later showed, this hypothesis did not hold.

Data Description

The dataset used in this project consisted of 4.5 million news headlines scraped from the top 10 news publishers between 2007 and 2022 [Krishnayah, J]. Each record included the headline text, publication source, date, and URL. The dataset was randomly split into training, validation, and test sets to ensure that model performance reflected generalization to unseen headlines.

The dataset included headlines from the following ten publishers:

- New York Times
- CNN
- FOX

- The Guardian
- Washington Post
- BBC
- Daily Mail
- USA Today
- CNBC
- New York Post

The average headline length across most publishers was approximately 10 words. Daily Mail headlines were a significant outlier, averaging 23 words per headline, which later proved to be an important factor in model performance.

Dataset Cleaning

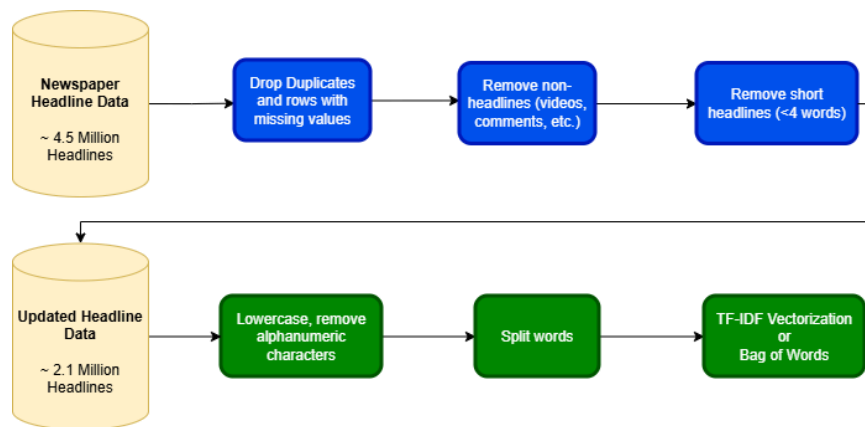


Figure 1. Dataset Cleaning

Significant preprocessing was required before training any models. The following steps were taken:

- Approximately 2.1 million duplicate URLs were removed to avoid repeated headlines.
- Roughly 30,000 duplicate headline texts were removed.
- Non-headline entries such as “Video,” “Comments,” and other non-news text were removed from the headline data.

These steps ensured that the dataset consisted only of unique, valid headlines and reduced noise that could negatively impact model performance. After filtering, the remaining headlines were cleaned, using case normalization and removing punctuation.

Methodology

Three different neural network architectures were implemented and evaluated, along with a traditional machine learning baseline. Each neural network included early stopping, to stop training once validation accuracy began to decrease.

Feed-Forward Neural Network (Bag-of-Words / TF-IDF)

Headlines were vectorized using Bag-of-Words or TF-IDF representations, producing large sparse feature vectors. These vectors were passed through a feed-forward neural network with dense layers and a softmax output. Despite its simplicity, this approach yielded the highest test accuracy of the models tested.

Long Short-Term Memory Network (LSTM)

For the LSTM model, headlines were tokenized into sequences and passed through an embedding layer, followed by an LSTM layer and a dense softmax output. This architecture was designed to capture word order and sequential patterns in text.

Convolutional Neural Network (1D CNN)

The CNN model used embeddings followed by one-dimensional convolutional layers and global max pooling. This architecture aimed to detect local n-gram patterns and stylistic phrases that might be characteristic of specific publishers.

Baseline Logistic Regression Model

A TF-IDF vectorizer combined with logistic regression was used as a baseline, achieving approximately 51% accuracy. All deep learning models were expected to outperform this baseline.

Results

Model	Test Accuracy
Bag-of-Words Feed Forward Network	59%
LSTM	58%
CNN	55%
Baseline (TF-IDF + Logistic Regression)	51%

Figure 2. Accuracy by Model

Deep learning models did not significantly outperform a simple feed-forward network using a bag-of-words. All models appeared to plateau in the same performance range of 55-59%, which was a slight improvement from the baseline logistic regression model.

Performance varied widely by publisher, with Daily Mail having a much higher test accuracy than other publishers. CNBC, which focuses more specifically on business news, had the second highest accuracy at 64.7%, indicating that the model may be identifying coverage differences when making classification predictions.

Publisher	Test Accuracy
Daily Mail	89.6%

CNBC	64.7%
New York Times	61.3%
BBC	61.3%
FOX	51.4%
Washington Post	47.8%
USA Today	47.1%
New York Post	47.1%
CNN	45.3%
The Guardian	40.0%

Figure 3. Accuracy by News Publisher

Evaluation and Error Analysis

Accuracy increased substantially with headline length, particularly for the CNN model. For headlines less than 10 words, accuracy remained around 40%; however, headlines greater than 15 words nearly doubled that accuracy, indicating that additional sentence context is important for distinguishing different publishers.

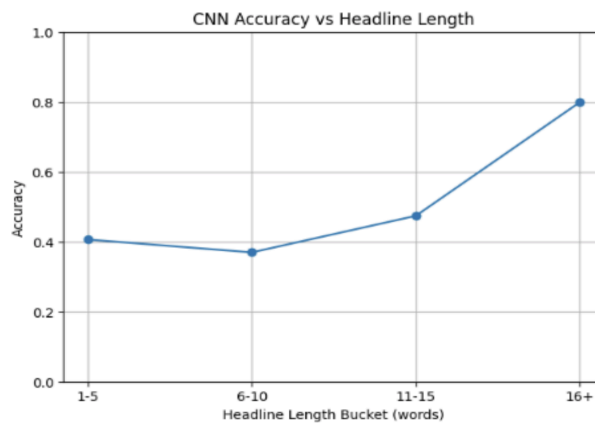


Figure 4. CNN Accuracy by Headline Length Bucket

Since Daily Mail headlines were generally 2-3 times longer than the other publishers, a greater number of words could be analyzed for Daily Mail headlines, leading to a much higher testing accuracy for Daily Mail.

Misclassification Patterns

While classification accuracy remained around 55-60% overall, certain publisher pairs had higher misclassification percentage than other pairs. This indicates that these outlets have similar tone, topic coverage, or geographic focus with one another, making it difficult for the model to discern between the two publishers.

Publisher 1	Publisher 2	Misclassification (1->2) %	Misclassification (2->1) %
FOX	New York Post	10.7%	13.0%
CNN	BBC	12.8%	10.5%
New York Times	Washington Post	10.3%	12.1%
The Guardian	BBC	10.3%	7.0%
FOX	Washington Post	8.8%	8.5%

Figure 5. Highest Misclassification Pairs

For example, New York Post was incorrectly classified as FOX 13% of the time, the highest misclassification pairing, indicating that these right-wing outlets have significant similarities in their headlines and coverage patterns. Additionally, the British outlets, BBC and The Guardian, had a high degree of misclassifications, likely from their greater focus on British and worldwide stories.

Discussion

The results indicate that headline length is a significant factor in classification accuracy. Deep learning models rely heavily on context and sequence information, which is limited in short headlines. Many outlets use similar wording for political and breaking news, making stylistic differences difficult to detect.

Bag-of-Words models were effective due to their reliance on keyword frequency rather than sequential context, which is more appropriate for short text. In contrast, LSTMs and CNNs did not have enough sequential information to leverage their strengths. Thus, these sequential models consistently failed to classify shorter, more generic headlines while longer, more descriptive headlines performed well. These trends continued with current day headlines that sometimes featured words unseen by the model.

A key limitation of this study is that it relies exclusively on headline text, which inherently restricts the amount of stylistic and contextual information available to the models. Deep learning would likely perform much better if lead paragraphs or full articles were included in the training and testing data.

Conclusion & Future Work

This project demonstrates that predicting a news publisher from headline text alone is possible, but inherently limited. The highest accuracy achieved was approximately 58–59%, and deep learning models (CNNs and LSTMs) did not outperform a simpler Feed Forward Network. With less than 60% accuracy, additional contextual information will be necessary if these models are to be implemented into source attribution.

Future work could include using longer text inputs, experimenting with lightweight transformer models, and incorporating stylistic features such as sentiment, punctuation usage, and reading level. Additionally, to minimize common misclassifications, model stacking could be implemented with one initial model determining a publisher pairing, and a subsequent model predicting among two similar publishers with a sigmoid activation function.

References

Krishnayah, J. (2023, July 17). *4.5m headlines from 2007-2022 [10 largest sites]*. Kaggle.
<https://www.kaggle.com/datasets/jordankrishnayah/45m-headlines-from-2007-2022-10-largest-sites>

Appendix 1: Contributions from Each Member

- **Nathan Ensley:** FFN Model training, visualizations, headline scraper for live demonstration
- **Nick Day:** Data cleaning, baseline modeling, CNN and LSTM model training
- **Christian Haskell:** Project organization, presentation design